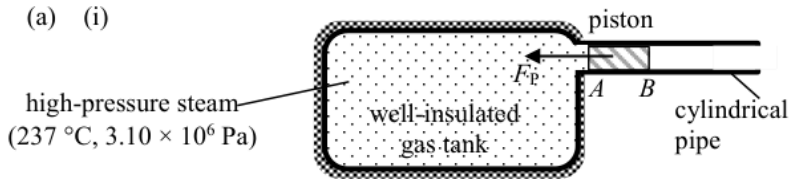


1.	(a)	$5 \times 0.02 \times 3300 \times (T - 4) = 0.60 \times 4200 \times (96 - T)$ $T = 85.347368 \text{ }^{\circ}\text{C} \approx 85.3 \text{ }^{\circ}\text{C}$	1M	Accept 85.0 °C to 85.4 °C [Marking Remark (MR): temperature difference must be correct & accept missing '5' – 1 M]
			1A	
	(b)	(i)	To compensate for / balance the heat loss (of the container with soup) to the surroundings.	2
				1A
		(ii)	$P \times 10 \times 60 = 2000 \times 9 + 16 \times 4200 \times 9$ $P = 1038 \text{ W} \approx 1040 \text{ W}$	1
				1M+1M
				1A
	(iii)	Smaller than 9 °C. As the temperature of (the container with) soup drops, the rate of heat loss to the surroundings becomes lower due to a smaller temperature difference (w.r.t. the surroundings).	Assume : Power of the heater = Rate of heat loss (of the container with soup) to the surroundings [MR: $P \times t$ ✓ + 1 term ✓ - 1M $P \times t$ ✓ + 2 term ✓ - 2M $P \times t$: accept $P \times 10$ ΔT : accept (96 – 9)]	3
				1A
				1A
				2

2. (a) (i)



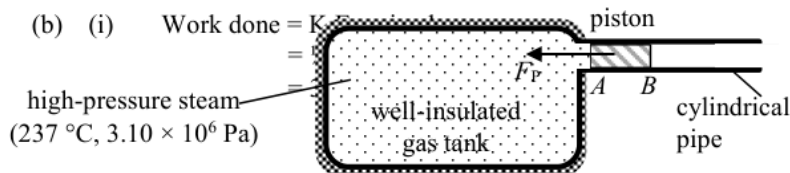
$$(ii) \quad F_P = (3.10 \times 10^6 - 1.0 \times 10^5) \times 0.67$$

$$= 2010000 \text{ N} = 2.01 \times 10^6 \text{ N}$$

$$(iii) \quad pV = nRT \Rightarrow V = \frac{nRT}{p}$$

$$V = \frac{(570/0.018)(8.31)(237 + 273)}{3.10 \times 10^6}$$

$$= 43.292419 \text{ m}^3 \approx 43.3 \text{ m}^3$$



$$(ii) \quad \text{Average acceleration } a = \frac{v - u}{t} = \frac{54 - 0}{1.5}$$

$$= 36 \text{ m s}^{-2}$$

(iii) Acceleration is decreasing (i.e. maximum at first). (According to kinetic theory,) once the steam expands, i.e. volume increases, its pressure decreases, thus the (pressure) force acting on the piston at A decreases, hence a smaller acceleration.

1A

 F_P correctly marked

F_P should be in contact with the piston.
Accept: no label of the force

1

1M

Accept: $F_P = (3.10 \times 10^6) \times 0.67$

1A

Accept $2.00 \times 10^6 \text{ N}$ to $2.01 \times 10^6 \text{ N}$

2

1M+1M

1st 1M find n ($570/0.018$)
2nd 1M $pV=nRT$, accept $T=237$

1A

Accept 43.0 m^3 to 43.3 m^3

3

1M

1A

Accept 37.9 MJ to 38.0 MJ

Accept: $a = 54/1.5 = 36 \text{ (m s}^{-2}\text{)}$
 $F = (2.6 \times 10^4) \times 36$
 $= 936000 \text{ (N)}$
 $s = 54^2 / (2 \times 36) = 40.5 \text{ (m)}$
 [or $v = at/2 = 36(1.5)/2$
 $= 27 \text{ (m s}^{-1}\text{)}$
 $W = Fs = 936000 \times 340.5$
 [or $W = Fvt = 936000 \times 27 \times 1.5$
 $= 37908000 \text{ (J)}$
 [1M+1A]

If $s = vt$ [0M 0A]

2

1M

1A

2

1st 1A for acceleration decreasing
 2nd 1A for volume increases/
length increases $\Rightarrow$
pressure decreases

1A

Alternative method:

The acceleration of the jet fighter is decreasing, 1A

As the air resistance (f) acts on the jet fighter increases with the speed. 1A

The net force ($ma = F - f$) is decreasing. 1A

3

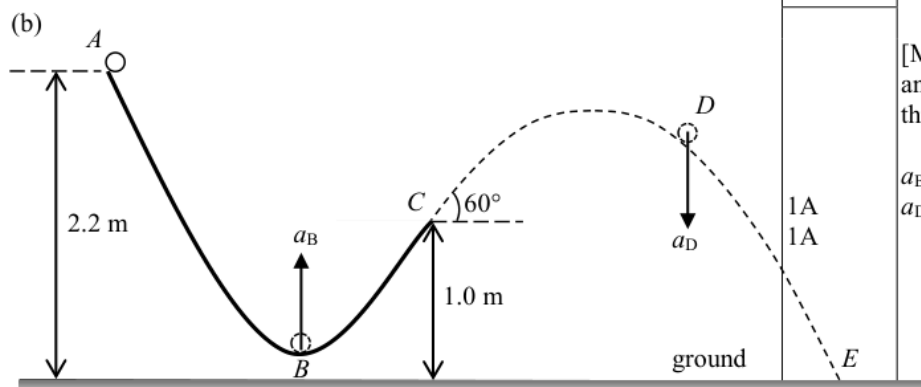
3.	(a)	With superconductor / extremely low or nearly zero resistance, a much larger current flows in the coils producing a stronger magnetic field (with less heat loss due to the current flow).	1A	[MR: low R + large I – 2A marks large I only – 1 A mark low R only – 0 A mark]					
			1A						
		2							
	(b)	P and Q : N (north pole) Repulsive force between like poles	1A						
			1A						
					2				
	(c)	(i)	As the train is not in contact with the rail, <u>there is no friction / interaction between the train and the rail</u> , thus smoother due to less vibration.		1A				
		(ii)	There is no friction / interaction between the train and the rail , and the propulsion of the train only needs to work against air resistance, therefore much faster.		1A				
			2						

4. (a) $v_B > v_C > v_D > v_A$

1A

[MR: Accept: $v_A < v_D < v_C < v_B$
Accept: B, C, D, A
NOT accept: A, D, C, B]

1



1A

1A

[MR: Accept the arrows drawn at anywhere on the vertical lines passing through the object at positions B & D]

a_B correctly marked
 a_D correctly marked

- (c) (i) Gravitational potential energy is changed to kinetic energy from A to B, and some kinetic energy is changed back to gravitational potential energy from B to C.

1A

2 [MR: 2 marks : All correct and complete: PE $\rightarrow$ KE & some KE $\rightarrow$ PE
1 mark :

- (i) PE $\rightarrow$ KE $\rightarrow$ PE or
(ii) Correctly describe only between 2 points (A $\rightarrow$ B / B $\rightarrow$ C / A $\rightarrow$ C)]

2

- (ii) $\frac{1}{2}mv^2 = mgh$
 $v^2 = 2(9.81)(2.2 - 1.0)$
 $v = 4.852216 \text{ m s}^{-1} \approx 4.85 \text{ m s}^{-1}$

1M

1A

[MR: h must be correct for 1M
Method using $v^2 - u^2 = 2as$ etc.
is NOT accepted]

For $g = 10 \text{ m s}^{-2}$,
 $v = 4.89898 \text{ m s}^{-1} \approx 4.90 \text{ m s}^{-1}$
Accept 4.85 m s^{-1} to 4.90 m s^{-1}

2

- (iii) Horizontal speed (at C) = $4.85 \times \cos 60^\circ$
 $= 2.426108 \text{ m s}^{-1} \approx 2.43 \text{ m s}^{-1}$

1M

$$\text{Distance} = 2.43 \times t = 2.55 \text{ m}$$

$$t = 1.051066 \text{ s} \approx 1.05 \text{ s}$$

1M

1A

Or

$$\text{Vertical speed (at C)} = 4.85 \times \sin 60^\circ$$

$$= 4.2002 \text{ m s}^{-1} \approx 4.20 \text{ m s}^{-1}$$

$$-1.0 = 4.20 \times t + \frac{1}{2}(-9.81)t^2$$

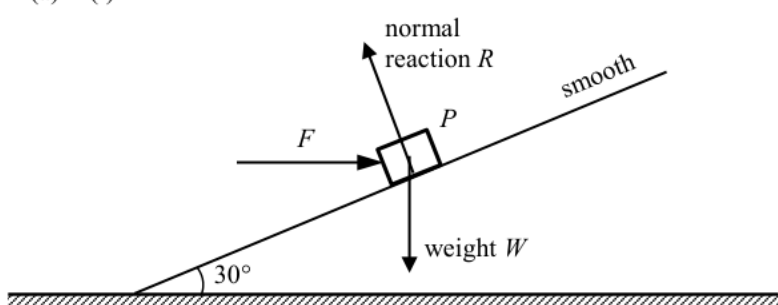
$$t = 1.051066 \text{ s} \approx 1.05 \text{ s}$$

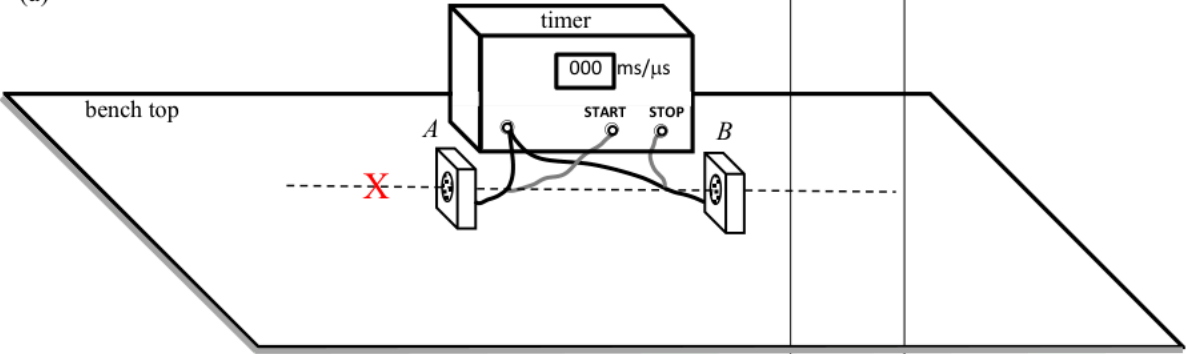
e.c.f. from (c)(ii)
For $g = 10 \text{ m s}^{-2}$,
Horizontal speed
 $= 2.44949 \text{ m s}^{-1} \approx 2.45 \text{ m s}^{-1}$
Vertical speed $\approx 4.24 \text{ m s}^{-1}$
and $t = 1.041033 \text{ s} \approx 1.04 \text{ s}$
Accept 1.04 s to 1.10 s

[MR:
1st M for correct expression for
initial horizontal /vertical speed

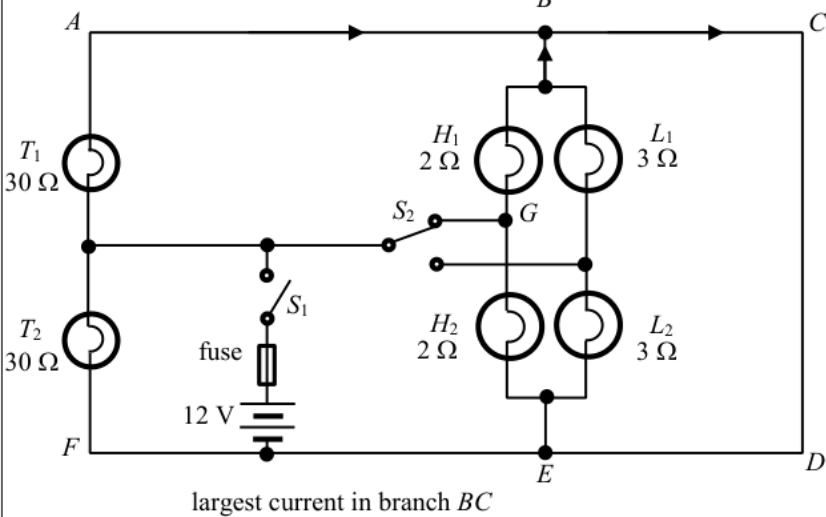
2nd M for correct method (equation) to
find t by considering horizontal
motion/ vertical motion

3

5. (a) (i)		1A 1A	Forces R (or N) and W (or Mg) correctly indicated and labelled. [MR: 1A for weight with correct label (W/Mg /weight/gravity) 1A for normal reaction with correct label (R/N /normal force/contact force/reaction) 1A only: if both W and R drawn (without/wrong label) Withhold 1A for additional force drawn (more than 2 forces) - W components added with correct labels (ignored) - W components added without correct labels (treated as additional force drawn)																					
		2																						
(ii) $R \cos 30^\circ = W = Mg$ and $R \sin 30^\circ = F$ $F = Mg \times \tan 30^\circ = 56.63806 \text{ N} \approx 56.6 \text{ N}$ $N = R = \frac{Mg}{\cos 30^\circ} = 113.27612 \text{ N} \approx 113 \text{ N}$ <u>Or</u> $R = W \cos 30^\circ + F \sin 30^\circ$ and $W \sin 30^\circ = F \cos 30^\circ$	1M 1A 1A (1M)	$M = 10 \text{ kg}$; $Mg = 98.1 \text{ N}$ [MR: 1M for both equations are correct] For $g = 10 \text{ m s}^{-2}$, $57.735027 \text{ N} \approx 57.7 \text{ N}$ $115.470054 \text{ N} \approx 115 \text{ N}$ [MR: 1M for both equations are correct]																						
		3																						
(b) (i) $g \sin \theta = 9.81 \sin 30^\circ = 4.905 \text{ m s}^{-2} \approx 4.91 \text{ m s}^{-2}$		1A	5 m s^{-2} for $g = 10 \text{ m s}^{-2}$,																					
		1																						
(ii) Decrease as the component of F perpendicular to the incline no longer acts on the block / incline <u>Or</u> as F is removed, the force pressing the block / incline would decrease (only the weight's component left)	1A 1A	<table><tr><th colspan="2" rowspan="2"></th><th colspan="3">Answer</th></tr><tr><th>De</th><th>In / Un</th><th>NA</th></tr><tr><td rowspan="3">Reason</td><td>✓</td><td>2</td><td>0</td><td>1</td></tr><tr><td>✗ / NA</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Not complete</td><td>1</td><td>0</td><td>0</td></tr></table>			Answer			De	In / Un	NA	Reason	✓	2	0	1	✗ / NA	0	0	0	Not complete	1	0	0	
		Answer																						
		De	In / Un	NA																				
Reason	✓	2	0	1																				
	✗ / NA	0	0	0																				
	Not complete	1	0	0																				
		2	NA – Not answered / Un - Unchanged																					

6. (a) 		
Correct location of 'X' (close to axis and on the side of the START microphone). Use a <u>metre rule</u> to measure the distance / separation D between the two microphones. Record the timer reading, which is the time interval Δt for the sound to travel from one microphone (START) to the other microphone (STOP), i.e. A to B.	1A 1A 1A	Accept ruler Any ONE of the measurements
(b) (i) Discard $539 \mu\text{s}$, $\Delta t = \frac{801 + 838 + 821}{3} = 820 \mu\text{s}$ The speed of sound in air $v = \frac{D}{\Delta t}$ $v = \frac{0.280}{820 \times 10^{-6}}$ $= 341.463415 \text{ m s}^{-1} \approx 341 \text{ m s}^{-1}$ (ii) Increase separation D between the microphones.	3 1M 1A 1A 3	1M for taking the average value of Δt and quoting the formula $v = \frac{D}{\Delta t}$ to calculate the speed of sound, where D is the separation between the microphones. Accept 340 m s^{-1} to 342 m s^{-1}

7.	(a)	(i)	Angle of incidence at A , $i_A = 90^\circ - 30^\circ = 60^\circ$	1A	
				1	
		(ii)	$n_g \sin c = n_c \sin 90^\circ$	1M	1M for correct equation with angle of refraction in the cladding = 90°
				1A	Accept 1.2 or $\frac{2}{\sqrt{3}}$
				2	
		(iii)	Total internal reflection. For light rays from O with $\theta > 30^\circ$, they will fail to undergo total internal reflection.	1A	Correct spelling
				1A	Accept $\theta \geq 30^\circ$
				2	
		(b)	(i)	1A	1A for 'taking different paths'
				1A	1A for 'different arrival time'
					NOT accept: Some of the energy (going at angles larger than 30° from the axis) would be lost due to leakage, thus lower intensity.
				2	
		(ii)	The refractive index of cladding should be increased, as $\frac{n_g}{n_c} = \frac{1}{\sin c}$, by <u>making $c / \sin c$ larger</u> , only those light rays close to the axis / within a smaller θ are totally internal reflected, (such that $\frac{n_g}{n_c}$ getting closer to 1).	1A	
				1A	0 A for the explanation if the choice for the change of n_c is INCORRECT.
					Note: More light rays will escape the optical fibre for a larger n_c .
				2	

8. (a) Because L_1 and L_2 (connected in series) are <u>shorted by BCDE</u>	1A	Accept p.d. across each low-beam headlight (L_1 or L_2) is 0.
	1	
(b) (i) p.d. across $T_2 = 12\text{ V}$	1A	
	1	
(ii)	1A	Any two currents correctly marked.
 largest current in branch BC	1A	
	1A	All currents correct.
(c) Power supplied $P = 2 \times \left(\frac{12^2}{30} + \frac{12^2}{2} \right)$ $= 153.6\text{ W} = 154\text{ W}$ (3 sig. fig.)	1A	
	3	
$\frac{V^2}{R_{eq}} = \frac{12^2}{R_{eq}} = 153.6$ (or $\frac{1}{R_{eq}} = \frac{1}{30} + \frac{1}{30} + \frac{1}{2} + \frac{1}{2}$)	1M	1M for same voltage across each bulb (T_1 , T_2 , H_1 and H_2) e.c.f. from (b)(i)
$R_{eq} = 0.9375\ \Omega \approx 0.938\ \Omega \approx 1\ \Omega$ (3 sig. fig.)	1A	
(d) max. current $I_{max} = \frac{V}{R_{eq}} = \frac{12}{0.9375} = 12.8\text{ A}$	1M	1M for all bulbs (T_1 , T_2 , H_1 and H_2) in parallel
(or $I_{max} = \frac{P}{V} = \frac{153.6}{12} = 12.8\text{ A}$)	1A	
Since the rating is slightly higher than the max. current (when high-beam headlights and taillights are switched on), it is suitable.	1M	e.c.f. from (c)
	1A	1A for 12.8 A and correct conclusion
	2	

9. (a)	Top view		1A	Accept: field lines are not on Figure 9.2
			1A	Correct direction (From A to B)
				Accept: field lines not in contact with the plates
				Perpendicular to plates, parallel and evenly spaced
				Accept: 3 or more field lines NOT accept: 2 or more mistakes in the drawing
			2	
(b)	(i)	$F \times d = (2.0 \times 10^{-5})(0.05 \cos 20^\circ)$ $= 9.396926 \times 10^{-7} \text{ N m} \approx 9.40 \times 10^{-7} \text{ N m}$	1M	1M $F \times d$
			1A	Accept: without consider the component / without component d (0.05, 0.05/2, 2×0.05 , 5, 5/2, 2×5) NOT accept: calculation using separation between plates d (0.1, 10)
			2	
	(ii)	$E = \frac{V}{d}$ $= \frac{5.0 \times 10^3}{0.1}$ $= 50\,000 \text{ V m}^{-1} \text{ or } \text{N C}^{-1} = 50 \text{ kV m}^{-1} \text{ or } \text{kN C}^{-1}$	1M	Accept: $\frac{5 \times 10^3}{0.1}$, $\frac{5 \times 10^3}{1}$, $\frac{5}{1}$, $\frac{5}{0.1}$
			1A	
			2	
	(iii)	$E = \frac{F}{Q}$ $Q = \frac{F}{E} = \frac{2.0 \times 10^{-5}}{5.0 \times 10^4}$ $= 4.0 \times 10^{-10} \text{ C}$	1M	e.c.f. from (b)(ii)
			1A	
			2	

10. (a) proton / ${}^1_1\text{H}$ / p / hydrogen nucleus	1A	Accept: proton / ${}^1_1\text{H}$ / p / 1_1p / H / hydrogen nucleus / hydrogen / hydrogen atom NOT accept: H_2
	1	
(b) Change in mass $\Delta m = (16.9947 + 1.0073) - (13.9993 + 4.0015)$ $= 0.0012 \text{ u}$ Energy = 0.0012×931 $= 1.1172 \text{ (MeV)} \approx 1.12 \text{ (MeV)}$ (Or = $0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13} \text{ J} \approx 1.79 \times 10^{-19} \text{ MJ}$)	1M 1A	1M for Δm Accept 1.10 to 1.12 (MeV) Answer should be in MeV, not J. MeV could be omitted [Energy = $0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13} \text{ J}$ $\approx 1.79 \times 10^{-19} \text{ MJ}$ $\approx 1.12 \text{ (MeV)}$] Accept $(1.79 \text{ to } 1.80) \times 10^{-19} \text{ MJ}$
	2	
(c) By the conservation of momentum, as before the reaction occurs the α particle has momentum, the total momentum of the products (= momentum of the α particle) must also be non-zero, i.e. the total KE of the products must greater than 0, thus the α particle should have a larger KE.	1A 1A	1st 1A should mention the product(s) has/have momentum 2nd 1A KE/speed of the products must greater than zero Accept: <u>kinetic</u> energy shared by other particles (NOT accept :energy shared.) NOT accept: against repulsion of nucleus / inelastic collisions / KE loss / consider binding energy [The kinetic energy of the α particle = energy calculated in (b) + total KE of the products]
	2	